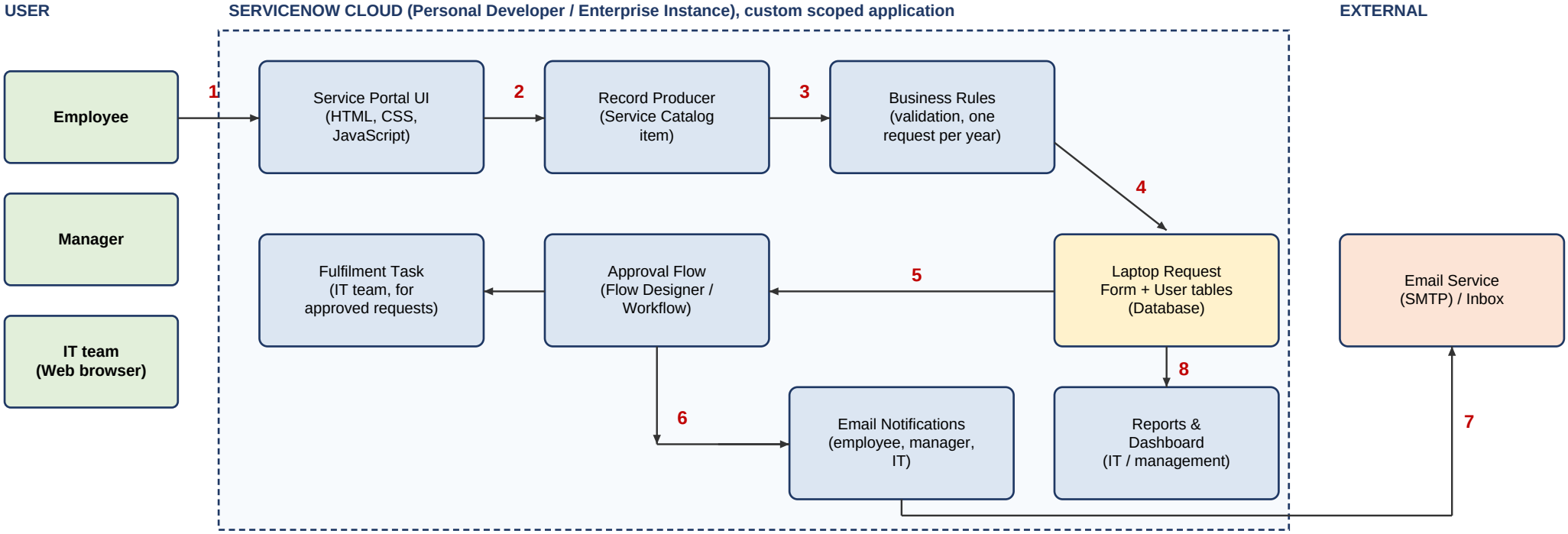
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	21 September 2026	**Student Name**	Pasapu Vineela
Team ID	Individual Project	**Regd. No.**	23KD1A05I0
Project Name	ServiceNow-Based Employee Laptop Request Application	**College**	Lendi Institute of Engineering and Technology
Maximum Marks	4 Marks		

Technical Architecture

Architecture of the ServiceNow-Based Employee Laptop Request Application, with the components and characteristics in Table-1 and Table-2. Example followed: Order processing during pandemics for offline mode. Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Flow: (1) User opens the Service Portal. (2) The Record Producer form collects the request. (3) Business rules validate it and enforce one request per employee per year. (4) The request is stored in the Laptop Request Form table with data from the User table. (5) The approval flow assigns the manager's approval task; approved requests create a fulfilment task for the IT team. (6) Notifications are triggered on submission and on the decision. (7) Email is delivered via the email service. (8) Reports and dashboards read from the table.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How the employee, manager and IT team interact with the application (web UI).	ServiceNow Service Portal: HTML, CSS, JavaScript, AngularJS widgets
2	Application Logic-1	Record Producer collects the laptop request and inserts it into the Laptop Request Form table.	ServiceNow Service Catalog (Record Producer), client and server scripts in JavaScript
3	Application Logic-2	Business rules for default values, validations and one-request-per-year restriction with duplicate error handling.	ServiceNow Business Rules (server-side JavaScript, GlideRecord)
4	Application Logic-3	Manager approval, status update, task assignment and email notifications.	Flow Designer / Workflow, Email Notifications
5	Database	Custom table Laptop Request Form (auto number, reference, choice fields); User table for requester and manager.	ServiceNow relational database tables
6	Cloud Database	Database service on the cloud instance that stores all application data.	ServiceNow instance database (cloud hosted)
7	File Storage	Attachments if a document needs to be added to a request.	ServiceNow attachment storage
8	External API-1	Email delivery for notifications to the employee, manager and IT.	Email service (SMTP) configured in the instance
9	External API-2	Not required in this phase; available for future integrations (for example an asset management system).	ServiceNow REST Table API (future)
10	Machine Learning Model	Not applicable in this phase.	Not applicable
11	Infrastructure (Server / Cloud)	Application deployment on the cloud instance, with configuration moved using Update Sets.	ServiceNow cloud (PDI or Enterprise instance)
	Local Server Configuration	Computer with at least 4 GB RAM, modern browser (Chrome, Edge or Firefox) and stable internet connection.	Local machine
	Cloud Server Configuration	ServiceNow Personal Developer Instance or Enterprise instance; custom application "Project Application" with Update Set.	ServiceNow cloud

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used by the Service Portal front end.	AngularJS, Bootstrap
2	Security Implementations	Role-based access control and ACLs, authenticated access, scoped application and secure browser access.	ServiceNow roles and ACLs, HTTPS/TLS, application scope
3	Scalable Architecture	Table-based, multi-user cloud platform with a separate presentation, logic and data layer (3-tier); can extend to approvals, notifications and reporting.	ServiceNow platform, scoped application
4	Availability	Cloud-hosted instance accessed through a browser, with availability managed by the platform provider.	ServiceNow cloud hosting
5	Performance	Lightweight form, server-side validations and indexed table lookups for the duplicate check; platform caching for portal pages.	ServiceNow business rules, table indexes, platform caching

References: https://c4model.com/ | https://www.ibm.com/cloud/architecture | https://aws.amazon.com/architecture